

Luiza Barros Reis Soezima

São Paulo, Brasil - <https://luizabrs.github.io/>

Education

[2024 - 2025]

Master's Degree – Cybersecurity / Grenoble INP, Ensimag – Grenoble, France

[2019 - 2024]

Bachelor's Degree - Computer Science / Universidade de São Paulo - USP, São Paulo – SP, Brazil

- Hackatons: SheHacks 2020, SheHacks 2022, Hackfools 2019
- [2019] MaratonUSP - Member of the coding marathon team focused on solving ICPC problems and participating in ICPC competitions.
- [2021, 2022] Student Council Representative - Representative of the students at the Council of Computer Science Department.
- [2023, 2024] Undergraduate Committee Student Representative - Representative of the students at the Undergraduate Committee.

[2014 – 2017] (interrupted)

Bachelor's Degree - Mechanical Engineering (interrupted) / Universidade Estadual de Campinas - UNICAMP, Campinas – SP, Brazil

[2011 - 2013]

High School / Colégio Bandeirantes, São Paulo – SP, Brazil

Courses

Summer Course – Scientific Initiation / IMPA – Instituto de Matemática Pura e Aplicada, Rio de Janeiro – RJ, Brazil

- Linear Algebra [2017], Real Analysis [2020], Introduction to Mathematical Economics [2020].

Course - General Business Program / Business Training Company, São Paulo – SP, Brazil

- [2023] Business training program focused on: Strategy, Corporate Finance, Valuation, Marketing and Negotiation.

Conferences and Advanced Schools

- [2023] INCACrypto 2023 - Introductory school in Cryptography / CriptoLatinos
 - [2023] ASCrypto 2023 - Advanced School in Cryptology / International Association for Cryptologic Research (IACR)
 - [2023] LatinCrypt 2023 - International Conference on Cryptology and Information Security in Latin America / International Association for Cryptologic Research (IACR)
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Academic Prizes and Honors

[2013]

London International Youth Science Forum / Imperial College London, Oxford University, Cambridge University, Cranfield University – England, United Kingdom

Represented the Brazilian youth scientists delegation as an honorable invitee. The LIYSF forum gathers around 500 leading young scientists from over 70 countries. The objective is to share and discuss how we see world changes and bring new sights and ideas back to our countries. In 2013, the theme was Science without Borders, which led to several discussions about the limits of the future of science.

[2015]

National Youth Science Forum / Australian National University, Canberra - Australia

Represented the Brazilian delegation as an honorable invitee. The NYSF is a forum that gathers students to potentialize their passion for science and technology. This forum combines the scientific field and academic experience to inspire students to choose different STEM careers.

[2018]

Helping Hand Challenge / UNICAMP, UCL, Royal Academy of Engineering

Developed a hand prosthesis adapted for low- and middle-income countries (LMIC), and won the Hand Challenge Contest, sponsored and guided by the Royal Academy of Engineering - UK and University College London, won best prosthesis in Brazil and second best worldwide.

[2019]

Hackfools / USP CodeLab, Facebook Inc.

Immersive hackathon that pushes students into innovative careers and stimulates the entrepreneurship spirit. 12-hour hackathon to foster creativity, innovation, group collaboration, and performance under pressure to bring a solution. Won an honorable mention with the PetMe application, a dating app for pets.

[2019 - 2020]

ICPC – International Collegiate Programming Contest / Universidade de São Paulo

“The International Collegiate Programming Contest is an algorithmic programming contest for college students. Teams of three, representing their university, work to solve the most real-world problems, fostering collaboration, creativity, innovation, and the ability to perform under pressure.”(<https://icpc.global/>)

- 2019 - Classified 5th on São Paulo regionals and 50th on Brazilian overall.

- 2020 - Classified 15th on São Paulo regionals and 57th on Brazilian overall.

[2020]

University Student Talent Award / CAPES - Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazilian Government

Awarded a scholarship from the education ministry (CAPES University Talent Award) for being one of the best Brazilian first-year students.

[2020]

SheHacks / USP CodeLab

SheHacks is a hackathon for university students who identify as women and LGBTQI+. That year, the theme was "Innovative ways of enhancing life quality and well-being of people". Companies such as Google, B2W, and iFood were the primary sponsors. Won honorable mention - 5th out of 21.

[2021]

Global Challenge Lab / Imperial College London, Tsinghua University, Huawei Inc., Nestlé Inc.

Selected among the candidate students to represent Brazil and the University of São Paulo.

The Global Challenge Lab is an entrepreneurship program partnership between Imperial College London Enterprise Lab and Tsinghua University's X-lab in Beijing. The objective is to create new ideas that help achieve UN SDG 3: Good Health and Wellbeing. The program includes workshops, guest speakers, and mentoring to help develop innovative ideas. For the Challenge, we worked with students from the University of Toronto, Imperial College, and Tsinghua University and developed a solution called Baby Beep, a voice tone pager using reused technological parts from previous pager manufacturers.

Work and Research Experience

[2014 - 2015]

Motriz EJ / Universidade Estadual de Campinas,

- Member of the junior enterprise company, finance administrative coordinator, ISO 9001:2008 internal auditor and quality manager. Organized the mechanical engineering week, with international companies and academic sponsors. Took part in workshops on finance, administration, and innovative business.

Skills: Leadership, Microsoft Excel, Strategic Planning, ISO9001, Negotiation, Project Management

[2015 - 2016]

InovEdu / Universidade Estadual de Campinas,

- Founded an educational NGO to develop the active learning method inside the university to develop students' learning methodology.

Skills: Social Communication, Cultural Sensitivity, Community Engagement, Advocacy and Networking

[March 2021 – March 2022]

Scientific Initiation – Machine Learning for Text Classification / Universidade de São Paulo, National Deputies Chamber, Justice Court of São Paulo

Supervision: Prof. André Carlos Ponce de Leon Ferreira de Carvalho

The project is a bilateral agreement between the University of São Paulo and the São Paulo State Justice Court (TJSP) to create an Artificial Intelligence that makes Natural Language Processing tasks on the Brazilian Legal court. My main work was to deal with machine learning techniques and do deep research on them and their applicability in our domain, testing and adapting them to enhance our results. My tasks are the following:

- Article mapping for machine learning techniques applied to named entity recognition.
- Article survey for BERT and Information Retrieval Techniques.
- Worked on inconsistency verification in the database that caused problems with preprocessing results.
- Working on Entities Annotation to create a new Legal Portuguese Corpus. Targeting something new and without any previous publication.
- Working on a revision article with other 3 PhD's of Named Entity Recognition applied in Portuguese and Legal Domain. It targets something new and without any previous publication.
- Working with an interdisciplinary team of politicians, computer engineers and computer scientists.

Scientific initiation with recognized scholarship.

Publication: UlyssesNER-Br: A Corpus of Brazilian Legislative Documents for Named Entity Recognition: https://link.springer.com/chapter/10.1007/978-3-030-98305-5_1

[Aug 2023 - PRESENT]

Scientific Initiation – Private Information Retrieval/ Universidade de São Paulo

Supervision: Prof. Hilder Vitor Lima Pereira

I aim to enhance the existing protocol FrodoPIR (<https://eprint.iacr.org/2022/981>), a stateful Private Information Retrieval (PIR) scheme rooted in the learning with errors (LWE) problem. My focus is on refining how the FrodoPIR communicates client-server data.

For the theoretical work, I had to delve into a few specific topics such as the GSW scheme, TFHE(Fast Fully Homomorphic Encryption over the Torus) and DGHV(Dijk-Gentry-Halevi-Vaikuntanathan) schemes, and the mathematical problems used to construct them, i.e., the LWE(Learning with errors) and AGCD(Approximate Greatest Common Divisor) problems.

Skills

Technical

- Programming Languages (ranked in proficiency): C, C++, Python, Octave/MATLAB, LISP, Java, JavaScript, Bash, Kotlin, Assembly
- Algorithms and Data Structures
- Text editing: Latex, HTML, CSS
- Tools: GNU/Linux, Docker, GCC
- Software Development Skills: Technical Writing, Test Driven Development, Agile, Kanban, Bug Report and fixing, Git

Language

- Spanish – Advanced – DELE Certificate - B2
- English – Advanced – Cambridge CAE Certificate - C1
- French – Upper-Intermediate – DELF Certificate – B2
- Japanese – Beginner
- Portuguese – Native